TIPalit: Product Requirements Document (PRD) & Improved Lean Canvas

Project Title: TIPalit: Market Viability and Consumer Adoption of a Verified Campus-Only Mobile Marketplace for Used Academic Goods and Student Trade
Institution: Technological Institute of the Philippines (TIP) – Quezon City & Manila
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Engineering Approach: Agentic Engineering / AI-Driven Full-Stack Mobile Development

1. Executive Summary & Product Overview

TIPalit is a mobile-first Consumer-to-Consumer (C2C) marketplace designed exclusively for students of the Technological Institute of the Philippines (TIP Quezon City and Manila campuses). It provides a centralized, verified digital platform where students can post, browse, buy, sell, and trade secondhand academic materials (such as textbooks, drafting tools, scientific calculators, uniforms, and learning gadgets).

Instead of relying on cluttered social media pages, unsafe public marketplaces, or expensive traditional e-commerce couriers, TIPalit brings curriculum-aligned discovery, in-app messaging, and safe on-campus meetup coordination into one mobile platform.

2. Improved Lean Canvas (Incorporating Strategic Enhancements)

2.1 Problem

1. **Cluttered Social Media Channels:** Buy-and-sell posts on the TIP Freedom Wall and college Facebook groups are quickly buried under confessions, memes, and announcements, making item discovery difficult.
2. **Safety & Scam Risks on Open Marketplaces:** Public platforms like Facebook Marketplace expose students to unverified strangers, fake accounts, bogus buyers/sellers, and unsafe off-campus meetups.
3. **High Costs & Delays in Traditional E-Commerce:** Platforms like Shopee and Lazada charge P50–P150 in shipping fees and take 3–5+ days for delivery—unnecessary friction when buyers and sellers walk the same campus halls.
4. **Lack of Academic Search Filters:** Existing platforms do not allow filtering by TIP subject codes (e.g., MATH 010, CIV 101), college departments, or specific campus locations.

5. **The Cold Start Vulnerability (Identified in Critique):** A closed-loop marketplace fails if early buyers log in to find empty shelves. Waitlisted buyer onboarding and pre-launch supply seeding are required.

2.2 Customer Segments

- **Primary Users:** Enrolled TIP undergraduate students from both Quezon City and Manila campuses (verified via `@tip.edu.ph`).
- **Buyers:** Budget-conscious students—especially incoming freshmen or students entering a new semester—looking for affordable secondhand textbooks, drafting tables/tools, scientific calculators, and PE/daily uniforms without shipping fees.
- **Sellers (Initial Seed Segment):** Upper-year, term-completing, and graduating students, along with academic student organizations, looking to declutter and recover cash/extra allowance from materials used in completed subjects.
- **B2B Advertisers & Sponsors:** Hyper-local campus-adjacent businesses (print shops, review centers, boarding houses/dorms, food stalls) and student-oriented digital/tech brands.

2.3 Unique Value Proposition (UVP)

"A verified campus marketplace exclusively for TIP students, where they can buy, sell, and trade academic materials more conveniently, without shipping fees or cluttered listings, and with added protection against scams."

- **100% Verified Community:** Restricted to `@tip.edu.ph` institutional accounts.
- **P0 Shipping Fees & Instant Same-Day Exchange:** Direct peer-to-peer meetups inside safe campus zones.
- **Curriculum-Aligned Discovery:** Organized by TIP course codes, departments, and item conditions.

2.4 Solution

1. **Institutional Verification Safeguard:** Mandatory registration and automated email verification using official `@tip.edu.ph` student accounts.
2. **Curriculum-Aligned Categorization & Search:** Instant filtering by subject code (e.g., MATH 010, CIV 101), college department, campus (QC or Manila), item condition, and price range.
3. **Sub-60-Second Listing Workflow:** Streamlined mobile upload flow for photos, condition tags, course codes, and pricing.
4. **Integrated In-App Messaging & Campus Meetup Coordinator:** Real-time chat with preset safe campus meetup locations (e.g., Library Plaza, Student Center, Library Lobby, Campus Cafeteria, Building Foyers) so students can trade between classes without sharing personal phone numbers or social media profiles.

5. **Community-Led Automated Moderation:** Two-way 5-star peer rating system and automated listing flag/report thresholds that hide suspicious posts without manual admin bottlenecks.

2.5 Channels & Go-To-Market (Cold Start Mitigation)

- **Phase 1: Targeted Supply Seeding (Pre-Launch / Sellers First):** Partner with graduating seniors, upper-year sections, and academic department organizations to pre-load high-demand inventory (calculators, drafting kits, textbooks, uniforms) *before* opening the app to general buyers.
- **Phase 2: Broad Campus Launch (Buyers & General Adoption):**
    - **Digital:** TIP Freedom Wall, college Facebook groups, and short-form TikTok video demonstrations.
    - **On-Campus:** Physical booths during freshman orientations, enrollment periods, and student fairs with QR codes for instant download and `@tip.edu.ph` verification.
    - **Institutional Partnerships:** Collaborations with the Supreme Student Council (SSC), department student councils, and student publications.

2.6 Revenue Streams (Hybrid Zoned Monetization Strategy)

Core peer-to-peer marketplace usage and standard listings remain **100% free** for students. Operating costs and sustainability are funded through a dual-phase hybrid model:

1. **Phase 1 (Day One) – Automated Affiliate Networks ("Student Perks & Deals" Tab):** Commission-based referral links via affiliate networks for tools and perks that genuinely benefit college students:
    - *Productivity & Study Tools:* Notion, Canva Pro, Grammarly.
    - *Tech & Developer Perks:* AWS sign-up/student credits, GitHub perks, Vercel hosting discounts, laptop/hardware retailer promos.
    - *Career & Upskilling:* Online courses (edX computer science/engineering tracks, interactive coding platforms like Mimo).
    - *Student Essentials:* Telco student data promos (Globe/Smart) and e-wallet cashback offers.
2. **Phase 2 (Post-Launch with DAU Leverage) – Hyper-Local B2B Direct Sponsorships (Homepage Banner):** Fixed-rate monthly sponsorship packages pitched using verified Daily Active User (DAU) metrics to businesses surrounding TIP QC and Manila:
    - Board exam review centers, nearby print/plotting shops, student dormitories/boarding houses, local cafes/food stalls, and companies recruiting TIP engineering and IT graduates.
3. **Optional Featured Listings (Micro-Transactions):** Optional P10–P20 fee for sellers who want to highlight or bump high-value items (e.g., graphing calculators, laptops, complete drafting sets) to the top of search results.

4. **Fallback Non-Intrusive Ad Network:** Native/banner ad slots via Google AdMob strictly as backfill when direct B2B sponsorship slots are unsold.

2.7 Cost Structure

- **Fixed Costs (Lean Cloud Architecture):**
  - **Backend-as-a-Service (Firebase):** P0 during launch and early growth using the Firebase Spark (Free) Plan for Authentication, Cloud Firestore, and Cloud Storage (aligning directly with the original Lean Canvas technology infrastructure proposal).
  - **App Store Developer Fees:** Google Play Store ($25 one-time fee) and Apple App Store ($99/year).
  - **Domain & Security:** Custom domain renewal and SSL certificates.
- **Variable Costs (Minimized via Automation):**
  - **Campus Promotions:** Physical flyers, tarpaulins, booth materials, and stickers during enrollment weeks.
  - **Escalation-Only Moderation:** Near-zero routine moderation costs due to automated community flagging and peer ratings; human admins only intervene on critical escalated disputes.

2.8 Key Metrics

1. **Supply Seeding Metrics (Pre-Launch):** Number of pre-registered upper-year sellers and active listings per department prior to buyer launch.
2. **User Acquisition:** Total app downloads and verified `@tip.edu.ph` account registrations.
3. **Platform Engagement:** Daily Active Users (DAU), Monthly Active Users (MAU), search-to-chat conversion rate, and active listings across subject codes.
4. **Transaction Success:** Confirmed campus meetups, completed exchanges (items marked as `sold`), and submitted two-way peer ratings.
5. **Monetization Metrics:** Affiliate link click-through/conversion rates in the "Student Perks" tab, direct B2B sponsorship retention, and optional P10–P20 boost purchases.

2.9 Unfair Advantage

- **Closed-Loop Verified Network Effect:** Exclusive `@tip.edu.ph` institutional gatekeeping creates high trust and hyper-relevant academic supply/demand that open platforms (Facebook Marketplace, Carousell, Shopee) cannot replicate. Every new verified TIP student increases both listing relevance and immediate on-campus liquidity.

3. Technical Architecture & Stack (For Agentic Engineering)

This specification is structured so AI coding agents can scaffold, implement, and test the full application modularly.

3.1 Tech Stack Overview

- **Mobile Frontend (Existing Codebase Integration):** Integrates directly into the existing mobile frontend UI without overwriting existing screens or styles, wiring UI components to a dedicated `src/services/firebase/` data layer.
- **Backend-as-a-Service (BaaS): Firebase**
    - **Firebase Authentication:** Email/Password authentication with mandatory email verification (`sendEmailVerification`) and `@tip.edu.ph` domain enforcement via Firestore Security Rules (`request.auth.token.email.matches('.*@tip[.]edu[.]ph$')`).
    - **Cloud Firestore:** NoSQL document database with real-time `onSnapshot` listeners for instant buyer-seller chat and composite indexes for curriculum-aligned filtering.
    - **Firebase Cloud Storage:** Dedicated storage buckets (`item_photos/` and `promo_assets/`) secured by Firebase Storage Security Rules.
    - **Cloud Functions for Firebase / Security Rules:** Automated triggers and rules for `@tip.edu.ph` validation, auto-hiding listings at `report_count >= 5`, and recalculating 5-star peer rating averages.
- **Web Admin / Analytics & Performance Monitoring:** Next.js / React admin & analytics dashboard deployed and monitored on **Vercel**, alongside **Firebase Analytics**.
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- **Version Control & Project Management: GitHub** (repository, CI/CD, agentic PR workflows) and **Notion** (sprint tracking & documentation).

4. Detailed Functional Requirements (FR)

FR-1: Institutional Onboarding & `@tip.edu.ph` Verification

- **FR-1.1:** Users must register with their full name, campus selection (`QC` or `Manila`), college department, and official `@tip.edu.ph` email address.
- **FR-1.2:** Client-side validation and Firebase Authentication with Firestore Security Rules must strictly reject any email address that does not end with `@tip.edu.ph`.
- **FR-1.3:** Upon registration, Firebase Authentication triggers `sendEmailVerification()` to send an automated verification link to the student's TIP email.
- **FR-1.4:** Unverified accounts are restricted to a "Verify Your Email" holding screen and cannot browse, post, or message until verification is completed.

FR-2: Curriculum-Aligned Listing Creation (Sub-60-Second Seller Flow)

- **FR-2.1:** Verified sellers can create a listing by uploading 1–5 photos (captured via camera or selected from gallery using `expo-image-picker`, compressed before uploading to Firebase Cloud Storage).
- **FR-2.2:** Sellers must input:
  - `title` (String, max 100 chars)
  - `description` (String, max 1000 chars)
  - `price` (Float, >= 0; 0 indicates free/trade-only)
  - `subject_code` (String/Tag, e.g., `MATH 010`, `CIV 101`, `GEN ED`, `PE`)
  - `category` (Enum: `Textbooks`, `Calculators`, `Drafting & Engineering Tools`, `Uniforms`, `IT & Gadgets`, `Other Supplies`)
  - `department` (Enum: e.g., `CCS / IT`, `CEA / Engineering & Architecture`, `CBE / Business`, `Arts`, `General`)
  - `condition` (Enum: `Brand New`, `Like New`, `Lightly Used`, `Well Used`)
  - `campus` (Enum: `TIP Quezon City`, `TIP Manila`)
- **FR-2.3:** Sellers can optionally toggle a "Featured Boost" (P10–P20 flow) to pin the item to the top of relevant searches.

FR-3: Search, Discovery & Academic Filtering (Buyer Flow)

- **FR-3.1:** The home feed displays active listings (`status == 'available'`) filtered by the user's default campus (`QC` or `Manila`), with boosted listings highlighted at the top.
- **FR-3.2:** Buyers can perform full-text searches on item titles/descriptions and combine filters for `subject_code`, `department`, `category`, `condition`, and `price` (min/max range).
- **FR-3.3:** Listing detail screens display item photos, full metadata, seller profile card (with verified badge, department, and average star rating), a "Message Seller" CTA, and a "Report Listing" action.

FR-4: Real-Time In-App Messaging & Campus Meetup Coordinator

- **FR-4.1:** Tapping "Message Seller" creates or opens a unique `chats` thread linking `buyer_id`, `seller_id`, and `listing_id`.
- **FR-4.2:** Messages are delivered in real time via **Cloud Firestore** `onSnapshot` **real-time listeners** (`chats/{chatId}/messages`).
- **FR-4.3:** Chat UI includes **Campus Meetup Quick-Chips** allowing either party to propose a structured meetup card containing:
  - **Designated Safe Location:** `Library Plaza`, `Student Center`, `Library Lobby`, `Campus Cafeteria`, or `Building Foyer`.
  - **Proposed Date & Time:** Aligned with class breaks.

    - **Payment Method Note:** `Cash on Meetup` or `E-Wallet (GCash/Maya) on Meetup`.
- **FR-4.4:** Sellers can change listing status directly from the chat header (`Available` → `Reserved` → `Sold`).

FR-5: Community-Led Moderation & Two-Way Peer Ratings

- **FR-5.1 (Two-Way Reviews):** Once a seller marks a listing as `Sold` to a specific buyer from the chat thread, both the buyer and seller are prompted to submit a 1–5 star rating and short feedback comment.
- **FR-5.2 (Automated Rating Calculation):** A Cloud Functions for Firebase trigger recalculates the user's `average_rating` and `total_reviews` on their `users_profile` document whenever a new review is created.
- **FR-5.3 (Automated Flagging & Hiding):** Any verified user can report a listing or chat user by selecting a reason (`Scam/Bogus`, `Inappropriate Content`, `Off-Campus/Prohibited Item`, `Harassment`).
- **FR-5.4 (Threshold Auto-Hide):** Each unique user report increments `report_count` on the target listing. A Cloud Functions for Firebase trigger automatically updates `listing.status` to `hidden` if `report_count >= 5`, removing it from public feeds immediately and flagging it in the Admin Console for human escalation review.

FR-6: Zoned Monetization ("Student Perks" & B2B Banner Slots)

- **FR-6.1 (Homepage B2B Sponsor Banner):** A non-intrusive, native-styled banner slot on the home feed driven by a `promotions` collection in Cloud Firestore (supporting local print shops, review centers, dorms, and university sponsors).
- **FR-6.2 (Dedicated "Student Perks & Deals" Tab):** A curated directory of student-beneficial affiliate deals categorized into:
    - *Study & Productivity* (Notion, Canva Pro, Grammarly)
    - *Tech & Cloud Perks* (AWS credits, GitHub, Vercel, hardware discounts)
    - *Upskilling & Courses* (edX, Mimo)
    - *Campus & Daily Essentials* (Telco data promos, e-wallet deals, local campus partner discounts)
- **FR-6.3:** Clicking any banner or affiliate perk logs an impression/click event in Cloud Firestore for sponsor reporting before opening the external tracking URL.

5. Firebase Backend Database Schema (Cloud Firestore) & Storage Specification

5.1 Cloud Firestore Collections

Collection 1: `users_profile`

Attribute	Type	Required	Indexed	Description
`user_id`	String	Yes	Unique	Matches Firebase Auth `uid`
`email`	String	Yes	Unique	Must end in `@tip.edu.ph`
`full_name`	String	Yes	No	Student's full name
`campus`	Enum	Yes	Yes	`QC`, `Manila`
`department`	String	Yes	Yes	e.g., `CCS`, `CEA`, `CBE`
`is_verified`	Boolean	Yes	Yes	Mirrors email verification state
`average_rating`	Float	No	No	Default `0.0` (1.0 to 5.0 scale)
`total_reviews`	Integer	No	No	Default `0`
`role`	Enum	Yes	No	`student`, `org_seller`, `admin`

Collection 2: `listings`

Attribute	Type	Type/Constraints	Indexed	Description
`seller_id`	String	Required	Yes	References `users_profile.user_id`
`title`	String	Max 100 chars	Fulltext	Item title
`description`	String	Max 1000 chars	Fulltext	Condition notes, edition, details
`price`	Float	Min 0	Yes	Price in PHP (`0` = Free/Swap)
`subject_code`	String	Max 20 chars	Yes	Normalized uppercase (e.g., `MATH 010`, `CIV 101`)

Attribute	Type	Type/Constraints	Indexed	Description
category	Enum	Required	Yes	Textbooks, Calculators, Drafting, Uniforms, Gadgets, Others
department	String	Required	Yes	Target college department
condition	Enum	Required	Yes	Brand New, Like New, Lightly Used, Well Used
campus	Enum	Required	Yes	QC, Manila
image_ids	String[]	Max 5 items	No	Array of Firebase Cloud Storage file paths (item_photos/ path)
status	Enum	Required	Yes	available, reserved, sold, hidden
is_boosted	Boolean	Default false	Yes	Paid P10–P20 visibility boost flag
boost_until	DateTime	Nullable	Yes	Expiration timestamp for boosted visibility
report_count	Integer	Default 0	Yes	Auto-hides listing when >= 5

Collection 3: chats

Attribute	Type	Required	Indexed	Description
listing_id	String	Yes	Yes	References listings.$id
buyer_id	String	Yes	Yes	References users_profile.us...
seller_id	String	Yes	Yes	References users_profile.us...
meetup_spot	String	No	No	e.g., Library Plaza, Student ...
meetup_time	DateTime	No	No	Agreed campus exchange t...
last_message	String	No	No	Preview snippet for chat in...

Attribute	Type	Required	Indexed	Description
updated_at	DateTime	Yes	Yes	Used for sorting inbox thre...

Collection 4: messages

Attribute	Type	Required	Indexed	Description
chat_id	String	Yes	Yes	References chats.$id
sender_id	String	Yes	No	References users_profile.user_id
content	String	Yes	No	Text body or structured meetup proposal
message_type	Enum	Yes	No	text, meetup_proposal, system_status
is_read	Boolean	Yes	No	Default false

Collection 5: reviews

Attribute	Type	Required	Indexed	Description
listing_id	String	Yes	Yes	References completed listings.$id
reviewer_id	String	Yes	Yes	Buyer or Seller ID
reviewee_id	String	Yes	Yes	Target user being rated
rating	Integer	Yes	No	1 to 5 stars
comment	String	No	No	Short transaction feedback

Collection 6: reports

Attribute	Type	Required	Indexed	Description
target_type	Enum	Yes	Yes	listing, user, message

Attribute	Type	Required	Indexed	Description
target_id	String	Yes	Yes	ID of reported listing or user
reporter_id	String	Yes	Yes	Prevents duplicate reports from same user
reason	Enum	Yes	No	Scam, Inappropriate, Prohibited Item, Harassment
details	String	No	No	Optional user explanation
status	Enum	Yes	Yes	pending, auto_hidden, resolved, dismissed

Collection 7: promotions (B2B Sponsors & Affiliate Perks)

Attribute	Type	Required	Indexed	Description
title	String	Yes	No	Brand or offer headline
placement	Enum	Yes	Yes	home_banner (B2B Direct), perks_tab (Affiliate)
category	Enum	Yes	Yes	Local Campus Biz, Study Tools, Tech & Cloud, Upskilling, Telco & Finance
campus_target	Enum	Yes	Yes	ALL, QC, Manila
image_id	String	Yes	No	Banner/Logo in Firebase Cloud Storage
target_url	URL	Yes	No	Affiliate tracking link or sponsor page
is_active	Boolean	Yes	Yes	Controls live visibility
clicks_count	Integer	No	No	Tracks engagement for B2B sponsor reports

1. ### 5.2 Firebase Cloud Storage Paths & Security Rules

2. **`item_photos/{userId}/{listingId}/{fileName}`:**
    - Allowed file extensions: `jpg`, `jpeg`, `png`, `webp`.
    - Max file size: `5 MB` per image.
    - Permissions: Restricted to verified `@tip.edu.ph` users (`request.auth != null && request.auth.token.email_verified == true && request.auth.token.email.matches('.*@tip[.]edu[.]ph$')`).
3. **`promo_assets/{fileName}`:**
    - Stores B2B sponsor banners and affiliate brand logos.
    - Permissions: Read for verified `@tip.edu.ph` users, Write restricted to admin role.

6. Agentic Engineering Implementation Roadmap

To build TIPalit efficiently using AI coding agents, execute the development in five deterministic phases:

- ### Milestone 1: Non-Destructive Firebase Integration into Existing Frontend
    - **Do NOT overwrite existing frontend UI/screens.** Preserve all current components, navigation, and styling.
    - Install Firebase SDK (`firebase` or `@react-native-firebase/app`) and initialize config in a standalone service module (`src/services/firebase.ts`).
    - Deploy `firestore.rules`, `firestore.indexes.json`, and `storage.rules` via the Firebase CLI.

Milestone 2: Institutional Auth & `@tip.edu.ph` Guardrails (Firebase Authentication)

- Build Onboarding, Login, Sign-Up, and Email Verification screens.
- Configure Firestore Security Rules and Cloud Functions for Firebase to guarantee only `@tip.edu.ph` emails can hold active sessions and access data.

Milestone 3: Core Marketplace (Listings, Storage & Curriculum Filters)

- Implement the Sub-60-Second Seller Upload flow wired to Firebase Cloud Storage and Cloud Firestore.
- Build the Home Feed and Search/Filter drawer querying indexed attributes (`subject_code`, `department`, `campus`, `condition`, `price`).

Milestone 4: Real-Time Chat, Meetup Coordinator & Community Moderation

- Build the Inbox and Chat Room screens powered by Cloud Firestore `onSnapshot` real-time listeners.
- Add the Campus Meetup Quick-Chip selector (`Library Plaza`, `Student Center`, etc.) and Seller Status toggles (`Available` / `Reserved` / `Sold`).
- Deploy Cloud Functions for Firebase triggers for `handle-report-threshold` (auto-hides listings at 5 reports) and `update-user-rating` (calculates 5-star peer averages upon transaction completion).

Milestone 5: Zoned Monetization, Analytics Dashboard & Supply Seeding Launch

- Build the "Student Perks & Deals" tab (populated with initial affiliate links) and the Homepage B2B Sponsor Banner component.
- Deploy a lightweight Admin & Analytics Dashboard on Vercel connected to Firebase to monitor the 3 Key Metrics (Verified Registrations, Active Listings/DAU, Completed Campus Exchanges) and review escalated moderation reports.
- Execute **Phase 1 Supply Seeding** with graduating seniors and student organizations prior to opening general buyer registration.